Radioactivity and particles

30 questions: 10 multiple choice, 20 structured.

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1 · Part 1

Paper 1 & 2 Units [1]

The activity of a radioactive sample falls from 6400 Bq to 800 Bq in 90 minutes.

What is the half-life of the sample in seconds?

A 30 s

B 1800 s

C 2700 s

D 5400 s

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2 · Part 1

Paper 1 & 2 Radioactivity [1]

The symbol $^{56}_{26}\text{Fe}$ represents a nucleus of iron.

What is the number of neutrons in this nucleus?

A 26

B 56

C 82

D 30

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3 · Part 1 Paper 1 & 2 Radioactivity [1]

Workers in a hospital department that uses radioactive sources wear badges that contain photographic film.

What is the purpose of the photographic film?

A It darkens when ionising radiation reaches it, showing how much radiation the worker has been exposed to.

B It absorbs ionising radiation so that the radiation cannot reach the worker's body.

C It emits radiation that cancels out the ionising radiation in the room.

D It blocks background radiation but lets through radiation from the hospital's sources.

4 · Part 1 Paper 1 & 2 Radioactivity [1]

In a hospital, one patient swallows a liquid that contains a radioactive isotope.

Separately, a sealed packet of food is placed in a beam of gamma rays from a source outside the packet, to kill bacteria.

Which row describes what has happened to the patient and to the food?

	patient	**food**
A	irradiated	irradiated
B	irradiated	contaminated
C	contaminated	irradiated
D	contaminated	contaminated

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5 · Part 1

Paper 1 & 2 | Radioactivity | [1]

An unstable nucleus emits a beta (β−) particle. The nucleus formed then emits an alpha (α) particle.

Which row gives the overall change in the atomic number and in the mass number?

	change in atomic number	change in mass number
A	−1	−4
B	−3	−4
C	−1	−5
D	+3	−4

6 · Part 1

Paper 1 & 2 | Radioactivity | [1]

Sample X has an activity of 1600 Bq and a half-life of 2.0 hours.

Sample Y has an activity of 400 Bq and a half-life of 6.0 hours.

Which row gives the activities of the two samples 6.0 hours later?

	activity of X in Bq	activity of Y in Bq
A	400	200
B	200	100
C	800	200
D	200	200

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7 · Part 1 Paper 1 & 2 Fission and fusion [1]

Which nuclear reactions can be a source of energy?

A fission only

B fission, fusion and radioactive decay

C fusion only

D fission and fusion only

8 · Part 1 Paper 1 & 2 Fission and fusion [1]

The core of a nuclear fission reactor is surrounded by a thick layer of concrete.

Why is the concrete used?

A to increase the pressure in the core so that fusion can take place

B to keep thermal energy in the core so that fission can continue

C to absorb ionising radiation so that it does not reach people outside the reactor

D to make the uranium fuel last longer by slowing down its radioactive decay

9 · Part 1 Paper 1 & 2 Fission and fusion [1]

When a nucleus of uranium-235 undergoes fission, energy is released.

In which form is this energy released?

A as chemical energy of the products

B as kinetic energy of the fission products

C as gravitational potential energy of the fission products

D as electrical energy stored in the fission products

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10 · Part 1

Paper 1 & 2

Fission and fusion

[1]

What is the energy source for stars?

A nuclear fusion

B nuclear fission

C combustion (burning) of hydrogen gas

D radioactive decay of heavy nuclei

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11 · Part 1

Paper 1 & 2 Radioactivity [6]

Potassium has two naturally occurring isotopes, $^{39}_{19}K$ and $^{40}_{19}K$. Potassium-40 is radioactive.

(a) State what is meant by the term isotopes. [2]

(b) Complete the table to show the number of protons and the number of neutrons in a nucleus of each isotope.

	number of protons	number of neutrons
$^{39}_{19}K$		
$^{40}_{19}K$		

[2]

(c) State the number of electrons in a neutral atom of potassium-40. [1]

(d) Calcium-40 is represented by $^{40}_{20}Ca$. Explain why $^{40}_{19}K$ and $^{40}_{20}Ca$ are not isotopes of the same element. [1]

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12 · Part 1 Paper 1 & 2 Radioactivity [6]

The sealed radioactive source in a school laboratory emits alpha (α) particles and gamma (γ) rays.

(a) Describe the nature of an alpha particle. [2]

(b) Describe the nature of a gamma ray. [1]

(c) Alpha particles from the source travel only a few centimetres in air. Gamma rays from the same source travel much further.

Explain, in terms of ionisation, why alpha particles have a much shorter range in air than gamma rays. [3]

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13 · Part 1

Paper 1 & 2 Radioactivity [7]

In an experiment, a student investigates the radiation emitted by a sealed radioactive source.

The student places a Geiger–Müller (GM) detector 3 cm from the source. The student measures the count rate with no absorber, and then with different absorbers placed between the source and the detector.

With the source removed, the detector records a background count rate of 30 counts per minute.

The table shows the count rates recorded by the detector with the source in place.

absorber	count rate in counts per minute
none	960
sheet of paper	410
3 mm of aluminium	405
2 cm of lead	150

(a) State the independent variable in this investigation. [1]

(b) Calculate the count rate due to the source alone when the 2 cm of lead is in place. [1]

(c) Deduce which types of radiation the source emits. Use the data in the table to justify your answer. [4]

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(d) State one precaution the student should take to reduce the risk from the source. [1]

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14 · Part 1

Paper 1 & 2 Radioactivity [7]

(a) Radium-226 decays by emitting an alpha particle. The equation for this decay is

$^{226}_{88}Ra \rightarrow {}^{A}_{Z}Rn + {}^{4}_{2}He$

Determine the values of A and Z. [2]

(b) Bismuth-214, $^{214}_{83}Bi$, decays by emitting a beta (β^-) particle to form a nucleus of polonium (symbol Po).

Give a balanced nuclear equation for this decay. [3]

(c) Some nuclei emit a gamma ray after an alpha or beta decay.

Describe the effect of emitting a gamma ray on the atomic number and the mass number of a nucleus. Give a reason for your answer. [2]

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15 · Part 1

Paper 1 & 2 Radioactivity [6]

(a) Some unstable nuclei emit a neutron.

Describe the effect of emitting a neutron on the atomic number and on the mass number of the nucleus. [2]

(b) Thorium-232, $^{232}_{90}Th$, decays in a series of steps to form radon-220, $^{220}_{86}Rn$. In this series, each step emits either one alpha particle or one beta (β−) particle. No other particles are emitted.

Determine the number of alpha particles and the number of beta particles emitted in the series. Show your working. [4]

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16 · Part 1 Paper 1 & 2 Radioactivity [6]

Background radiation is present everywhere. It comes from sources on Earth and from space.

(a) Describe two sources of background radiation on Earth. [2]

(b) State one source of background radiation from space. [1]

(c) A teacher uses a Geiger–Müller detector to measure the background count in a laboratory. No radioactive sources are present. The counts recorded in five separate one-minute intervals are

18, 23, 20, 25, 19

Calculate the mean background count rate. Give the unit. [2]

(d) Nothing in the laboratory changed while the readings were taken. Explain why the five readings are different. [1]

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17 · Part 1 Paper 1 & 2 Radioactivity [7]

The table shows how the activity of a sample of a radioactive isotope changes with time.

time in min	activity in Bq
0	720
2	509
4	360
6	255
8	180
10	127

(a) State what is meant by the half-life of a radioactive isotope. [2]

(b) Use the data in the table to determine the half-life of this isotope. [2]

(c) Predict the activity of the sample at a time of 16 minutes. [2]

(d) Explain why the activity of the sample decreases with time. [1]

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18 · Part 1

Paper 1 & 2 Radioactivity [6]

In a laboratory, a technician uses a detector to measure the count rate from a radioactive source.

The background count rate is 30 counts per minute.

At the start, the detector reads 1630 counts per minute with the source in place.

The half-life of the source is 3.0 hours.

(a) Calculate the count rate due to the source alone at the start. [1]

(b) Calculate the count rate the detector will read 12 hours later, with the source still in place. [3]

(c) Determine how long after the start the detector reading will fall to 55 counts per minute. [2]

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19 · Part 1

Paper 1 & 2 Radioactivity [5]

Iodine-131 has a half-life of 8 days.

One sample contains 6.4×10^{14} undecayed nuclei of iodine-131.

(a) Calculate the number of undecayed iodine-131 nuclei left after 24 days. [2]

(b) Calculate the number of iodine-131 nuclei that decay during these 24 days. [1]

(c) One student says, "After 16 days, all of the iodine-131 in the sample will have decayed."

Explain why the student is wrong. [2]

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20 · Part 1 Paper 1 & 2 Radioactivity [7]

Technetium-99m is used as a medical tracer. It is injected into a patient, and a detector outside the patient's body detects the radiation it emits. Technetium-99m emits gamma rays and has a half-life of 6 hours.

(a) Explain why a tracer that emits gamma rays is used rather than one that emits alpha particles. [3]

(b) Explain why a half-life of 6 hours is suitable for a tracer of this kind. [2]

(c) Describe one other use of radioactivity in medicine. [2]

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21 · Part 1 Paper 1 & 2 Radioactivity [7]

In a paper factory, the paper passes between a radioactive source that emits beta (β−) particles, placed above the paper, and a detector placed below the paper.

The count rates in this question have been corrected for background radiation.

When the paper has the correct thickness, the count rate is 450 counts per minute.

The count rate is used by a computer to adjust the gap between two rollers that press the paper. A smaller gap makes the paper thinner.

(a) Explain why a source of beta particles is used, rather than a source of alpha particles or a source of gamma rays. [3]

(b) The count rate rises from 450 counts per minute to 520 counts per minute.

Deduce whether the paper has become thicker or thinner, and state how the computer should change the gap between the rollers. [2]

(c) Explain why using a source with a short half-life would cause the paper to become too thin over time. [2]

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22 · Part 1 Paper 1 & 2 Radioactivity [7]

People who work with radioactive materials must protect themselves from ionising radiation.

(a) Describe the difference between contamination and irradiation. [2]

(b) Explain why a source that emits alpha particles is more dangerous inside the body than outside the body. [3]

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(c) Describe two ways in which ionising radiation can harm living organisms. [2]

23 · Part 1 Paper 1 & 2 Radioactivity [7]

Radioactive waste from hospitals, industry and nuclear power stations must be disposed of safely.

(a) One isotope in some radioactive waste is caesium-137, which has a half-life of 30 years.

Calculate the time taken for the activity of a sample of caesium-137 to fall to one-sixteenth (1/16) of its starting value. [2]

(b) Describe the problems that arise from the disposal of radioactive waste, and how the risks can be reduced. [5]

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24 · Part 1

Paper 1 & 2 Fission and fusion [5]

The equation shows one way in which a nucleus of uranium-235 can undergo fission.

$^{1}_{0}n + {}^{235}_{92}U \rightarrow {}^{141}_{56}Ba + {}^{92}_{36}Kr + x\,{}^{1}_{0}n$

(a) Determine the value of x. Show your working. [2]

(b) Describe how this reaction can lead to a chain reaction in a large piece of uranium-235. [3]

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25 · Part 1

Paper 1 & 2 Fission and fusion [7]

In a nuclear reactor, fuel rods containing uranium-235 are surrounded by a moderator. Control rods can be raised out of, or lowered into, the reactor core.

(a) Explain the role of the moderator. [2]

(b) Explain how the control rods are used to control the rate of fission. [2]

(c) An engineer uses a computer model of a reactor. The table shows the number of fissions in each generation of the chain reaction. After generation 3, the position of the control rods was changed.

generation	number of fissions
1	1000
2	1000
3	1000
4	1100
5	1210
6	1331

Determine the factor by which the number of fissions increases from one generation to the next after generation 3. Use your answer to predict the number of fissions in generation 7. [2]

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(d) State how the control rods should now be moved to return the reactor to a steady rate of fission. [1]

26 · Part 1 Paper 1 & 2 Fission and fusion [5]

The equation shows a nuclear fusion reaction between two isotopes of hydrogen.

$^{2}_{1}H + ^{3}_{1}H \rightarrow ^{4}_{2}He + X$

(a) Determine the mass number and the atomic number of X, and identify X. [3]

(b) The total mass of the products is less than the total mass of the two hydrogen nuclei.

Explain what this shows about the reaction. [2]

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27 · Part 1 Paper 1 & 2 Fission and fusion [5]

Scientists are building experimental reactors in which hydrogen nuclei fuse. Inside these reactors, the hydrogen must be at a temperature of more than 100 million °C and at a high pressure.

(a) Explain why nuclear fusion does not happen at low temperatures and pressures. [4]

(b) Suggest one reason why it is difficult to build a power station that uses nuclear fusion. [1]

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28 · Part 1 Paper 1 & 2 Fission and fusion [5]

Nuclear fission and nuclear fusion are two types of nuclear reaction.

(a) Explain the difference between nuclear fission and nuclear fusion. [4]

(b) State one way in which nuclear fission and nuclear fusion are similar. [1]

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29 · Part 1

Paper 1 & 2 Fission and fusion [6]

In a lesson, a teacher uses a model to show how a chain reaction works.

Many mousetraps are set on a large tray. Each mousetrap has two table-tennis balls resting on it. When a mousetrap springs, it throws its two balls into the air, and they land on other mousetraps.

The teacher starts the model by dropping one ball onto one mousetrap.

(a) State what the table-tennis balls represent and what the mousetraps represent in this model. [2]

(b) The first mousetrap to spring is stage 1. The two mousetraps it sets off are stage 2, and so on.

Assume that every ball lands on a mousetrap that has not yet sprung. Calculate the number of mousetraps that spring in stage 5. [2]

(c) The teacher sets up the model again. This time, sheets of card are held above part of the tray, so that some of the balls land on the card instead of on a mousetrap.

State the part of a nuclear reactor that the sheets of card represent, and explain your answer. [2]

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30 · Part 1 Paper 1 & 2 Fission and fusion [6]

Each fission of a uranium-235 nucleus releases 3.2×10^{-11} J of energy.

One sample of nuclear fuel contains 2.5×10^{21} nuclei of uranium-235.

(a) Calculate the total energy released if every uranium-235 nucleus in the sample undergoes fission. [2]

(b) Burning 1.0 kg of coal releases 3.0×10^{7} J of energy.

Calculate the mass of coal that would release the same amount of energy as the sample of nuclear fuel. [2]

(c) After it has been used in a reactor, the fuel is more dangerous to handle than new fuel.

Explain why. [2]

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